

Lecture 6: Balanced Search Trees

Jessica Sorrell

September 17, 2026

601.433/633 Introduction to Algorithms

Slides by Mike Dinitz

Announcements

- ▶ HW1 due now, HW2 released
 - ▶ Oral presentations!
 - ▶ Form group of 3 (required!) in Canvas, sign up for presentation slot as a group in Canvas.
 - ▶ Format: HW has three questions. Grader will randomly assign each student to present one question. Rest of group can chime in occasionally.
- ▶ Regrade policy: 120 hours (five days) from when grades released
 - ▶ Don't abuse this!
 - ▶ If too many of your regrade requests do not result in positive changes, will ban you from regrade requests
 - ▶ Grading can go down!

Introduction

Today, and next few weeks: data structures.

- ▶ Since “Data Structures” a prereq, focus on advanced structures and on interesting analysis

Introduction

Today, and next few weeks: data structures.

- ▶ Since “Data Structures” a prereq, focus on advanced structures and on interesting analysis

Today and later: data structures for *dictionaries*

Introduction

Today, and next few weeks: data structures.

- ▶ Since “Data Structures” a prereq, focus on advanced structures and on interesting analysis

Today and later: data structures for *dictionaries*

Definition

A *dictionary data structure* is a data structure supporting the following operations:

- ▶ **insert(key,object)**: insert the (key, object) pair.
- ▶ **lookup(key)**: return the associated object
- ▶ **delete(key)**: remove the key and its object from the data structure. We may or may not care about this operation.

Obvious Approaches

Reminder: all running times for *worst case*

Obvious Approaches

Reminder: all running times for *worst case*

Approach 1: Sorted array

Obvious Approaches

Reminder: all running times for *worst case*

Approach 1: Sorted array

- ▶ Lookup:

Obvious Approaches

Reminder: all running times for *worst case*

Approach 1: Sorted array

- ▶ Lookup: $O(\log n)$

Obvious Approaches

Reminder: all running times for *worst case*

Approach 1: Sorted array

- ▶ Lookup: $O(\log n)$
- ▶ Insert:

Obvious Approaches

Reminder: all running times for *worst case*

Approach 1: Sorted array

- ▶ Lookup: $O(\log n)$
- ▶ Insert: $\Omega(n)$

Obvious Approaches

Reminder: all running times for *worst case*

Approach 1: Sorted array

- ▶ Lookup: $O(\log n)$
- ▶ Insert: $\Omega(n)$

Approach 2: Unsorted (linked) list

Obvious Approaches

Reminder: all running times for *worst case*

Approach 1: Sorted array

- ▶ Lookup: $O(\log n)$
- ▶ Insert: $\Omega(n)$

Approach 2: Unsorted (linked) list

- ▶ Insert:

Obvious Approaches

Reminder: all running times for *worst case*

Approach 1: Sorted array

- ▶ Lookup: $O(\log n)$
- ▶ Insert: $\Omega(n)$

Approach 2: Unsorted (linked) list

- ▶ Insert: $O(1)$

Obvious Approaches

Reminder: all running times for *worst case*

Approach 1: Sorted array

- ▶ Lookup: $O(\log n)$
- ▶ Insert: $\Omega(n)$

Approach 2: Unsorted (linked) list

- ▶ Insert: $O(1)$
- ▶ Lookup:

Obvious Approaches

Reminder: all running times for *worst case*

Approach 1: Sorted array

- ▶ Lookup: $O(\log n)$
- ▶ Insert: $\Omega(n)$

Approach 2: Unsorted (linked) list

- ▶ Insert: $O(1)$
- ▶ Lookup: $\Omega(n)$

Obvious Approaches

Reminder: all running times for *worst case*

Approach 1: Sorted array

- ▶ Lookup: $O(\log n)$
- ▶ Insert: $\Omega(n)$

Approach 2: Unsorted (linked) list

- ▶ Insert: $O(1)$
- ▶ Lookup: $\Omega(n)$

Goal: $O(\log n)$ for both.

Obvious Approaches

Reminder: all running times for *worst case*

Approach 1: Sorted array

- ▶ Lookup: $O(\log n)$
- ▶ Insert: $\Omega(n)$

Approach 2: Unsorted (linked) list

- ▶ Insert: $O(1)$
- ▶ Lookup: $\Omega(n)$

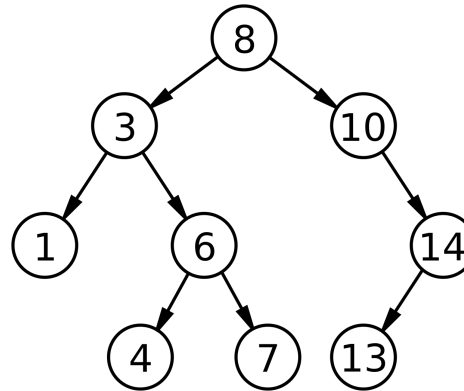
Goal: $O(\log n)$ for both.

Approach today: search trees

Binary Search Tree Review

Binary search tree:

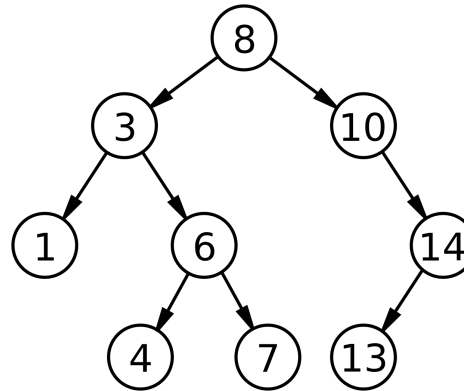
- ▶ All nodes have at most **2** children
- ▶ Each node stores (key, object) pair
- ▶ All descendants to left have smaller keys
- ▶ All descendants to the right have larger keys



Binary Search Tree Review

Binary search tree:

- ▶ All nodes have at most **2** children
- ▶ Each node stores (key, object) pair
- ▶ All descendants to left have smaller keys
- ▶ All descendants to the right have larger keys



Lookup: follow path from root!

Dictionary Operations in Simple Binary Search Tree

insert(x):

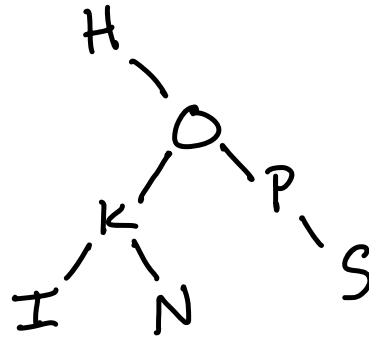
- ▶ If tree empty, put x at root
- ▶ Else if $x < \mathbf{root.key}$ recursively insert into left child
- ▶ Else (if $x > \mathbf{root.key}$) recursively insert into right child

Dictionary Operations in Simple Binary Search Tree

insert(x):

- ▶ If tree empty, put x at root
- ▶ Else if $x < \text{root.key}$ recursively insert into left child
- ▶ Else (if $x > \text{root.key}$) recursively insert into right child

Example: H O P K I N S
- -



Simple Binary Search Tree: Analysis

Pluses: easy to implement

Simple Binary Search Tree: Analysis

Pluses: easy to implement

(Worst-case) Running time:

Simple Binary Search Tree: Analysis

Pluses: easy to implement

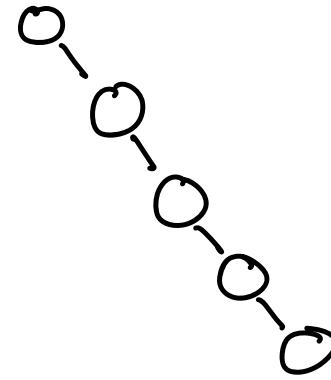
(Worst-case) Running time: if depth d , then $\Theta(d)$

Simple Binary Search Tree: Analysis

Pluses: easy to implement

(Worst-case) Running time: if depth d , then $\Theta(d)$

- ▶ If very unbalanced d could be $\Omega(n)$!



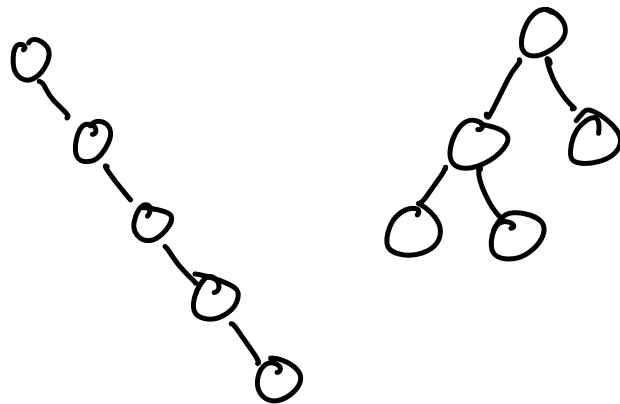
Simple Binary Search Tree: Analysis

Pluses: easy to implement

(Worst-case) Running time: if depth d , then $\Theta(d)$

- ▶ If very unbalanced d could be $\Omega(n)$!

Want to make tree *balanced*.



Simple Binary Search Tree: Analysis

Pluses: easy to implement

(Worst-case) Running time: if depth d , then $\Theta(d)$

- ▶ If very unbalanced d could be $\Omega(n)$!

Want to make tree *balanced*.

Rest of today:

- ▶ B-trees: perfect balance, not binary
- ▶ Red-black trees: approximate balance, binary
- ▶ Turn out to be related!

B-Trees

B-tree Definition

Parameter $t \geq 2$.

B-tree Definition

Parameter $t \geq 2$.

Definition (B-tree with parameter t)

1. Each node has between $t - 1$ and $2t - 1$ keys in it (except the root has between 1 and $2t - 1$ keys). Keys in a node are stored in a sorted array.
2. Each non-leaf has degree (number of children) equal to the number of keys in it plus 1. If v is a node with keys $[a_1, a_2, \dots, a_k]$ and the children are $[v_1, v_2, \dots, v_{k+1}]$, then the tree rooted at v_i contains only keys that are greater than a_{i-1} and less than a_i (except the edge cases: the tree rooted at v_1 has keys less than a_1 , and the tree rooted at v_{k+1} has keys greater than a_k).
3. All leaves are at the same depth.

B-tree Definition

Parameter $t \geq 2$.

Definition (B-tree with parameter t)

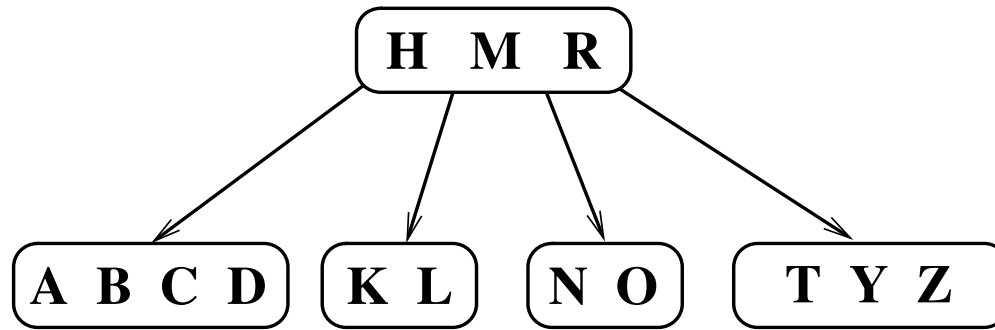
1. Each node has between $\underline{t - 1}$ and $\underline{2t - 1}$ keys in it (except the root has between $\mathbf{1}$ and $\mathbf{2t - 1}$ keys). Keys in a node are stored in a sorted array.
2. Each non-leaf has degree (number of children) equal to the number of keys in it plus $\mathbf{1}$. If $\mathbf{v}$ is a node with keys $[\mathbf{a_1}, \mathbf{a_2}, \dots, \mathbf{a_k}]$ and the children are $[\mathbf{v_1}, \mathbf{v_2}, \dots, \mathbf{v_{k+1}}]$, then the tree rooted at $\mathbf{v_i}$ contains only keys that are greater than $\mathbf{a_{i-1}}$ and less than $\mathbf{a_i}$ (except the edge cases: the tree rooted at $\mathbf{v_1}$ has keys less than $\mathbf{a_1}$, and the tree rooted at $\mathbf{v_{k+1}}$ has keys greater than $\mathbf{a_k}$).
3. All leaves are at the same depth.

When $t = 2$ known as a *2-3-4 tree*, since $\#$ children either 2, 3, or 4

B-tree: Example

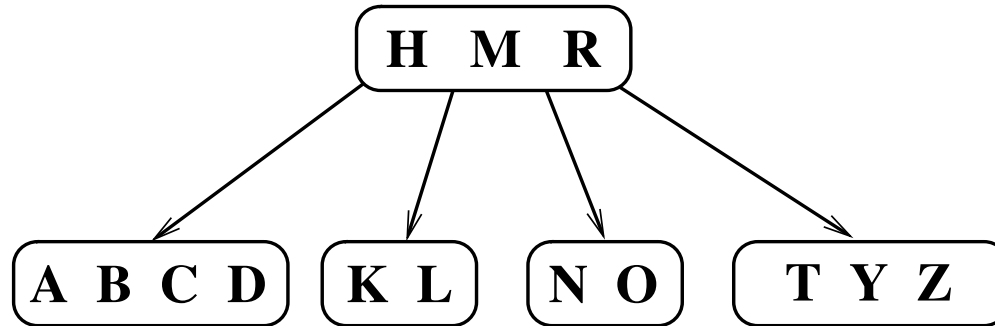
$t = 3$:

- ▶ Root has between **1** and **5** keys, non-roots have between **2** and **5** keys
- ▶ Non-leaves have between **3** and **6** children (root can have fewer).

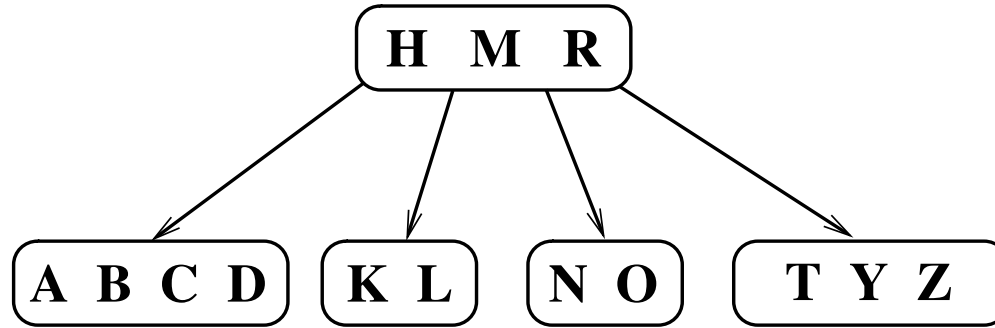


Lookups

Binary search in array at root. Finished if find item, else get pointer to appropriate child, recurse.



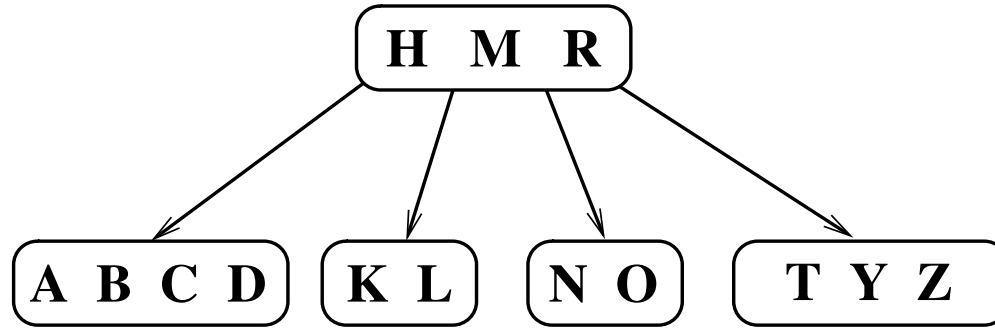
Insert(*x*)



Obvious approach: do a lookup, put *x* in leaf where it should be.

- ▶ Example: insert ***E***

Insert(x)

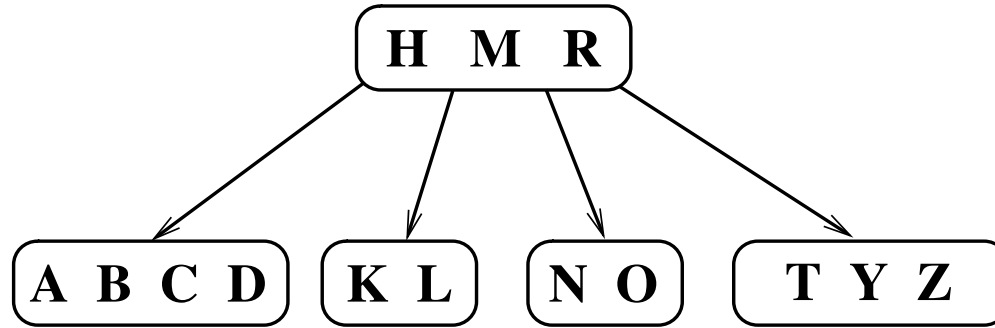


Obvious approach: do a lookup, put x in leaf where it should be.

► Example: insert ***E***

Problem: What if leaf is *full* (already has $2t - 1$ keys)?

Insert(x)



Obvious approach: do a lookup, put x in leaf where it should be.

- ▶ Example: insert **E**

Problem: What if leaf is *full* (already has $2t - 1$ keys)?

Split:

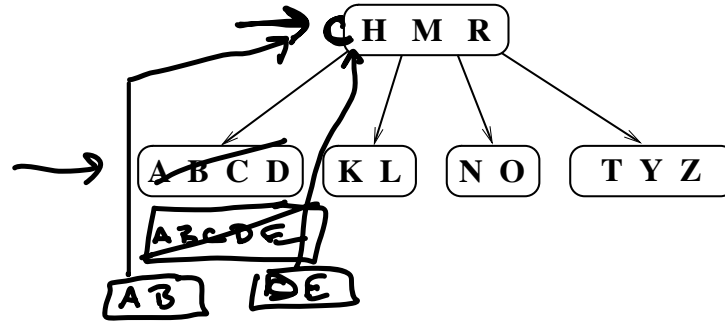
- ▶ Only used on full nodes (nodes with $2t - 1$ keys) whose parents are not full.
- ▶ Pull median of its keys up to its parent
- ▶ Split remaining $2t - 2$ keys into two nodes of $t - 1$ keys each. Reconnect appropriately.

Insert (continued)

Insert: do a lookup and insert at leaf, but when we encounter a full node on way down, split it.

Insert (continued)

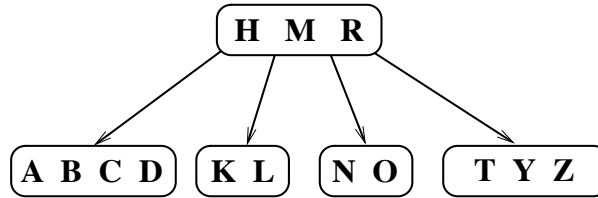
Insert: do a lookup and insert at leaf, but when we encounter a full node on way down, split it.



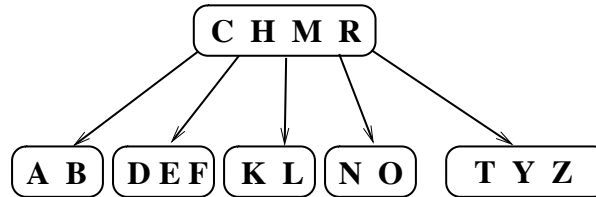
Insert **E**, **F** into example.

Insert (continued)

Insert: do a lookup and insert at leaf, but when we encounter a full node on way down, split it.

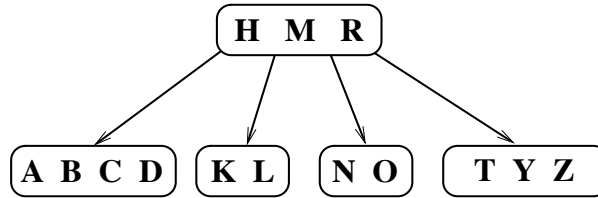


Insert **E**, **F** into example.

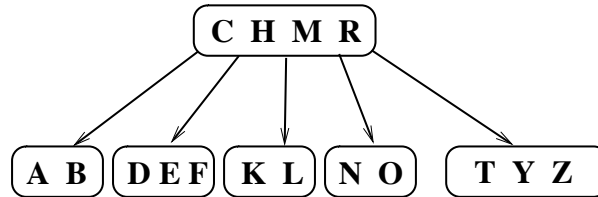


Insert (continued)

Insert: do a lookup and insert at leaf, but when we encounter a full node on way down, split it.

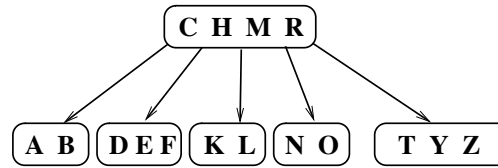


Insert **E**, **F** into example.

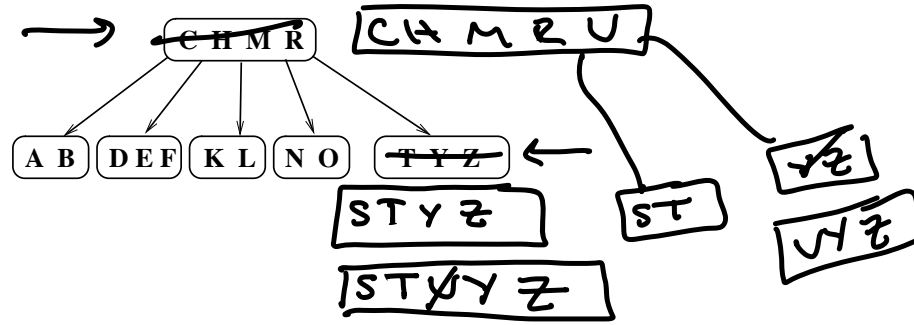


Note: since split *on the way down*, when a node is split, its parent is not full!

Example continued

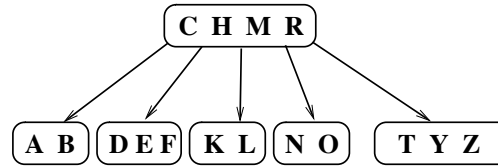


Example continued

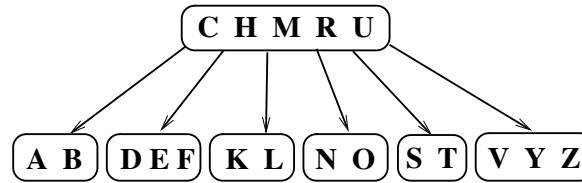


Insert **S**, **U**, **V**:

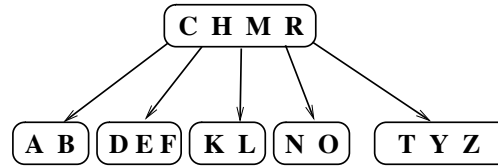
Example continued



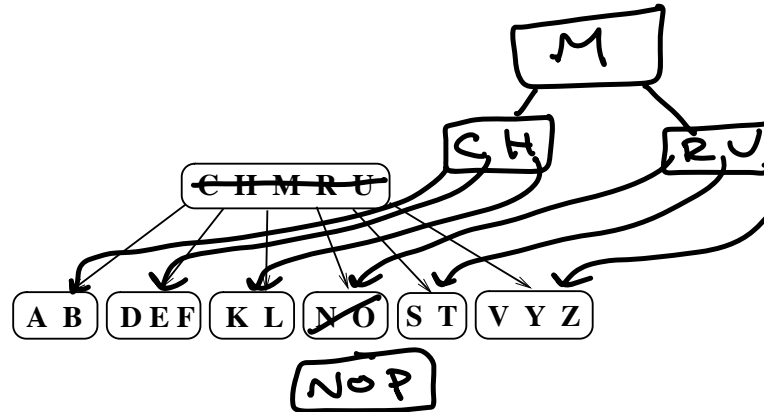
Insert ***S***, ***U***, ***V***:



Example continued

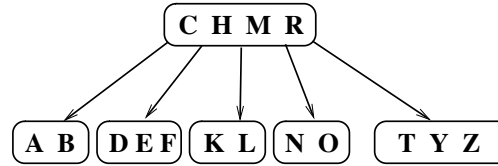


Insert **S**, **U**, **V**:

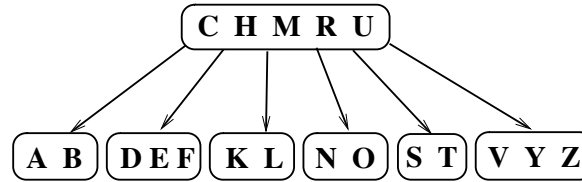


Insert **P**:

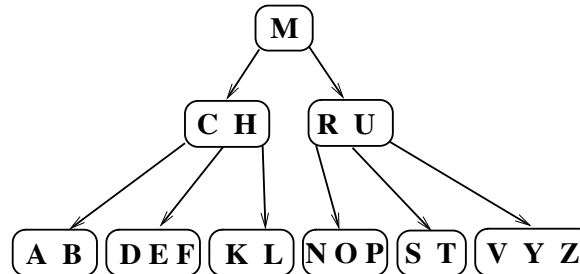
Example continued



Insert ***S***, ***U***, ***V***:



Insert ***P***:



Insert: Correctness sketch

Induction. Start with a valid B-tree, insert x .

Insert: Correctness sketch

Induction. Start with a valid B-tree, insert x .

Third property (all leaves at same depth):

Insert: Correctness sketch

Induction. Start with a valid B-tree, insert x .

Third property (all leaves at same depth): Tree grows up. ✓

Insert: Correctness sketch

Induction. Start with a valid B-tree, insert x .

Third property (all leaves at same depth): Tree grows up. ✓

First property (all nodes other than root have between $t - 1$ and $2t - 1$ keys):

Insert: Correctness sketch

Induction. Start with a valid B-tree, insert x .

Third property (all leaves at same depth): Tree grows up. ✓

First property (all nodes other than root have between $t - 1$ and $2t - 1$ keys):

- ▶ No split:

Insert: Correctness sketch

Induction. Start with a valid B-tree, insert x .

Third property (all leaves at same depth): Tree grows up. ✓

First property (all nodes other than root have between $t - 1$ and $2t - 1$ keys):

- ▶ No split: only leaf changes, was not full (or would have split)

Insert: Correctness sketch

Induction. Start with a valid B-tree, insert x .

Third property (all leaves at same depth): Tree grows up. ✓

First property (all nodes other than root have between $t - 1$ and $2t - 1$ keys):

- ▶ No split: only leaf changes, was not full (or would have split)
- ▶ Split:

Insert: Correctness sketch

Induction. Start with a valid B-tree, insert x .

Third property (all leaves at same depth): Tree grows up. ✓

First property (all nodes other than root have between $t - 1$ and $2t - 1$ keys):

- ▶ No split: only leaf changes, was not full (or would have split)
- ▶ Split: Parent was not full. New nodes have exactly $t - 1$ keys.

Insert: Correctness sketch

Induction. Start with a valid B-tree, insert x .

Third property (all leaves at same depth): Tree grows up. ✓

First property (all nodes other than root have between $t - 1$ and $2t - 1$ keys):

- ▶ No split: only leaf changes, was not full (or would have split)
- ▶ Split: Parent was not full. New nodes have exactly $t - 1$ keys.

Second property (correct degrees, subtrees have keys in correct ranges):

Insert: Correctness sketch

Induction. Start with a valid B-tree, insert x .

Third property (all leaves at same depth): Tree grows up. ✓

First property (all nodes other than root have between $t - 1$ and $2t - 1$ keys):

- ▶ No split: only leaf changes, was not full (or would have split)
- ▶ Split: Parent was not full. New nodes have exactly $t - 1$ keys.

Second property (correct degrees, subtrees have keys in correct ranges): Hooked nodes up correctly after split. ✓

B-tree running time

Suppose n keys, depth d

B-tree running time

Suppose n keys, depth $d \leq O(\log_t n)$

B-tree running time

Suppose n keys, depth $d \leq O(\log_t n)$

Lookup:

- ▶ Binary search on array in each node we pass through

B-tree running time

Suppose n keys, depth $d \leq O(\log_t n)$

Lookup:

- ▶ Binary search on array in each node we pass through $\implies O(\log t)$ time per node.

B-tree running time

Suppose n keys, depth $d \leq O(\log_t n)$

Lookup:

- ▶ Binary search on array in each node we pass through $\implies O(\log t)$ time per node.
- ▶ Total time $O(\underline{d} \times \underline{\log t}) = O(\underline{\log_t n} \times \log t) = O(\underline{\frac{\log n}{\log t}} \times \underline{\log t}) = O(\log n)$

B-tree running time

Suppose n keys, depth $d \leq O(\log_t n)$

Lookup:

- ▶ Binary search on array in each node we pass through $\implies O(\log t)$ time per node.
- ▶ Total time $O(d \times \log t) = O(\log_t n \times \log t) = O(\frac{\log n}{\log t} \times \log t) = O(\log n)$

Insert:

B-tree running time

Suppose n keys, depth $d \leq O(\log_t n)$

Lookup:

- ▶ Binary search on array in each node we pass through $\implies O(\log t)$ time per node.
- ▶ Total time $O(d \times \log t) = O(\log_t n \times \log t) = O(\frac{\log n}{\log t} \times \log t) = O(\log n)$

Insert:

- ▶ Same as lookup, but need to split on the way down & insert into leaf

B-tree running time

Suppose n keys, depth $d \leq O(\log_t n)$

Lookup:

- ▶ Binary search on array in each node we pass through $\implies O(\log t)$ time per node.
- ▶ Total time $O(d \times \log t) = O(\log_t n \times \log t) = O(\frac{\log n}{\log t} \times \log t) = O(\log n)$

Insert:

- ▶ Same as lookup, but need to split on the way down & insert into leaf
- ▶ Total: lookup time + splitting time + time to insert into leaf

B-tree running time

Suppose n keys, depth $d \leq O(\log_t n)$

Lookup:

- ▶ Binary search on array in each node we pass through $\implies O(\log t)$ time per node.
- ▶ Total time $O(d \times \log t) = O(\log_t n \times \log t) = O(\frac{\log n}{\log t} \times \log t) = O(\log n)$

Insert:

- ▶ Same as lookup, but need to split on the way down & insert into leaf
- ▶ Total: lookup time + splitting time + time to insert into leaf
 - ▶ Insert into leaf:

B-tree running time

Suppose n keys, depth $d \leq O(\log_t n)$

Lookup:

- ▶ Binary search on array in each node we pass through $\implies O(\log t)$ time per node.
- ▶ Total time $O(d \times \log t) = O(\log_t n \times \log t) = O(\frac{\log n}{\log t} \times \log t) = O(\log n)$

Insert:

- ▶ Same as lookup, but need to split on the way down & insert into leaf
- ▶ Total: lookup time + splitting time + time to insert into leaf
 - ▶ Insert into leaf: $O(t)$

B-tree running time

Suppose n keys, depth $d \leq O(\log_t n)$

Lookup:

- ▶ Binary search on array in each node we pass through $\implies O(\log t)$ time per node.
- ▶ Total time $O(d \times \log t) = O(\log_t n \times \log t) = O(\frac{\log n}{\log t} \times \log t) = O(\log n)$

Insert:

- ▶ Same as lookup, but need to split on the way down & insert into leaf
- ▶ Total: lookup time + splitting time + time to insert into leaf
 - ▶ Insert into leaf: $O(t)$
 - ▶ Splitting time:

B-tree running time

Suppose n keys, depth $d \leq O(\log_t n)$

Lookup:

- ▶ Binary search on array in each node we pass through $\implies O(\log t)$ time per node.
- ▶ Total time $O(d \times \log t) = O(\log_t n \times \log t) = O(\frac{\log n}{\log t} \times \log t) = O(\log n)$

Insert:

- ▶ Same as lookup, but need to split on the way down & insert into leaf
- ▶ Total: lookup time + splitting time + time to insert into leaf
 - ▶ Insert into leaf: $O(t)$
 - ▶ Splitting time: $O(t)$ per split

B-tree running time

Suppose n keys, depth $d \leq O(\log_t n)$

Lookup:

- ▶ Binary search on array in each node we pass through $\implies O(\log t)$ time per node.
- ▶ Total time $O(d \times \log t) = O(\log_t n \times \log t) = O(\frac{\log n}{\log t} \times \log t) = O(\log n)$

Insert:

- ▶ Same as lookup, but need to split on the way down & insert into leaf
- ▶ Total: lookup time + splitting time + time to insert into leaf
 - ▶ Insert into leaf: $O(t)$
 - ▶ Splitting time: $O(t)$ per split $\implies O(td) = O(t \log_t n)$ total

B-tree running time

Suppose n keys, depth $d \leq O(\log_t n)$

Lookup:

- ▶ Binary search on array in each node we pass through $\implies O(\log t)$ time per node.
- ▶ Total time $O(d \times \log t) = O(\log_t n \times \log t) = O(\frac{\log n}{\log t} \times \log t) = O(\log n)$

Insert:

- ▶ Same as lookup, but need to split on the way down & insert into leaf
- ▶ Total: lookup time + splitting time + time to insert into leaf
 - ▶ Insert into leaf: $O(t)$
 - ▶ Splitting time: $O(t)$ per split $\implies O(td) = O(t \log_t n)$ total
- ▶ $O(t \log_t n)$ = $O(\frac{t}{\log t} \log n)$ total

B-tree notes

Used a lot in databases

- ▶ Large t : shallow trees. Fits well with memory hierarchy

B-tree notes

Used a lot in databases

- ▶ Large t : shallow trees. Fits well with memory hierarchy

$t = 2$:

- ▶ 2-3-4 tree
- ▶ Can be implemented as *binary* tree using *red-black trees*

Red-Black Trees

Red-Black Trees: Intro

B-Trees great, but binary is nice: lookups very simple!
Want *binary* balanced tree.

Red-Black Trees: Intro

B-Trees great, but binary is nice: lookups very simple!

Want *binary* balanced tree.

- ▶ Classical and super important data structure question
- ▶ Many solutions!

Red-Black Trees: Intro

B-Trees great, but binary is nice: lookups very simple!

Want *binary* balanced tree.

- ▶ Classical and super important data structure question
- ▶ Many solutions!

Most famous: *red-black trees*

- ▶ Default in Linux kernel, used to optimize Java HashMap, ...
- ▶ Today: Quick overview, connection to 2-3-4 trees.
- ▶ *Not* traditional or practical point of view on red-black trees. See book!

2-3-4 trees to binary

Can we turn a 2-3-4 tree into a binary tree with all the same properties?

2-3-4 trees to binary

Can we turn a 2-3-4 tree into a binary tree with all the same properties?

- ▶ *No*: can't have perfect balance!

2-3-4 trees to binary

Can we turn a 2-3-4 tree into a binary tree with all the same properties?

- ▶ *No*: can't have perfect balance!
- ▶ Just need depth $O(\log n)$

2-3-4 trees to binary

Can we turn a 2-3-4 tree into a binary tree with all the same properties?

- ▶ *No*: can't have perfect balance!
- ▶ Just need depth $O(\log n)$

Nodes in 2-3-4 tree have degree 2, 3, or 4

2-3-4 trees to binary

Can we turn a 2-3-4 tree into a binary tree with all the same properties?

- ▶ *No*: can't have perfect balance!
- ▶ Just need depth $O(\log n)$

Nodes in 2-3-4 tree have degree 2, 3, or 4

- ▶ Degree 2: good!

2-3-4 trees to binary

Can we turn a 2-3-4 tree into a binary tree with all the same properties?

- ▶ *No*: can't have perfect balance!
- ▶ Just need depth $O(\log n)$

Nodes in 2-3-4 tree have degree 2, 3, or 4

- ▶ Degree 2: good!
- ▶ Degree 4:

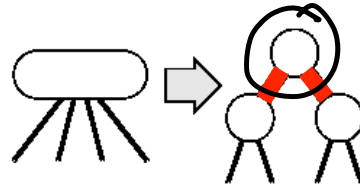
2-3-4 trees to binary

Can we turn a 2-3-4 tree into a binary tree with all the same properties?

- ▶ *No*: can't have perfect balance!
- ▶ Just need depth $O(\log n)$

Nodes in 2-3-4 tree have degree 2, 3, or 4

- ▶ Degree 2: good!
- ▶ Degree 4:



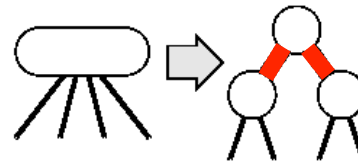
2-3-4 trees to binary

Can we turn a 2-3-4 tree into a binary tree with all the same properties?

- ▶ No: can't have perfect balance!
- ▶ Just need depth $O(\log n)$

Nodes in 2-3-4 tree have degree 2, 3, or 4

- ▶ Degree 2: good!
- ▶ Degree 4:



- ▶ Degree 3:

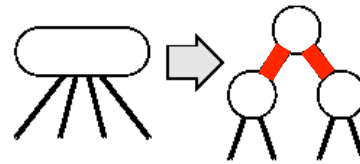
2-3-4 trees to binary

Can we turn a 2-3-4 tree into a binary tree with all the same properties?

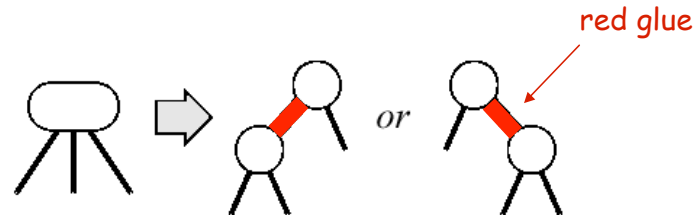
- ▶ *No*: can't have perfect balance!
- ▶ Just need depth $O(\log n)$

Nodes in 2-3-4 tree have degree 2, 3, or 4

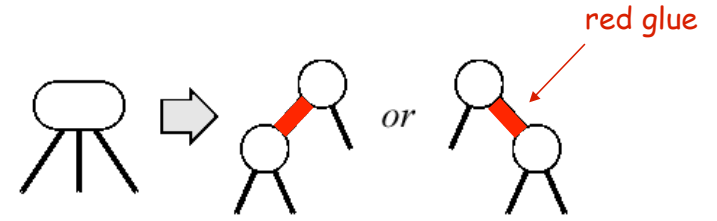
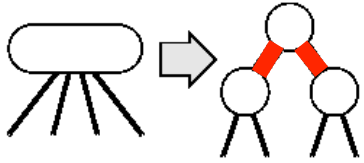
- ▶ Degree 2: good!
- ▶ Degree 4:



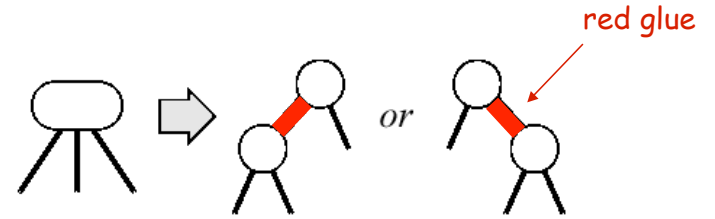
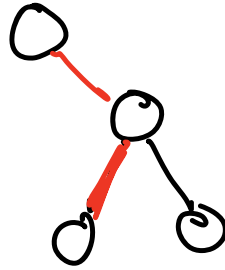
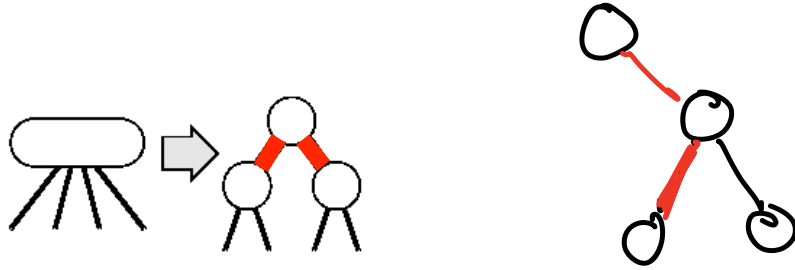
- ▶ Degree 3:



Important Properties



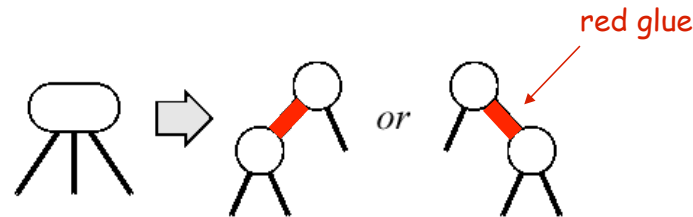
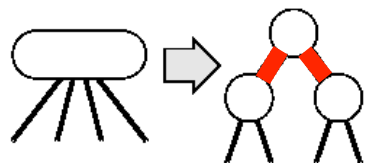
Important Properties



1. Never have two red edges in a row.

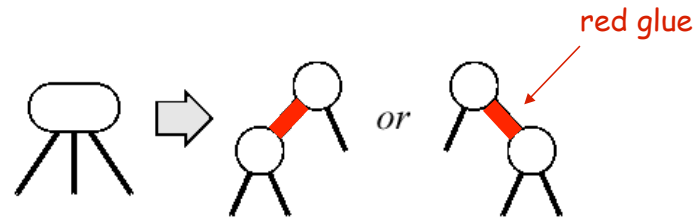
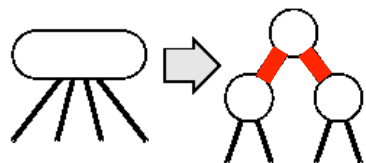
- ▶ Red edge is “internal”, never have more than one “internal” edge in a row.

Important Properties



1. Never have two red edges in a row.
 - ▶ Red edge is “internal”, never have more than one “internal” edge in a row.
2. Every leaf has same number of *black* edges on path to root (*black-depth*)
 - ▶ Each black edge is a 2-3-4 tree edge
 - ▶ All leaves in 2-3-4 tree at same distance from root

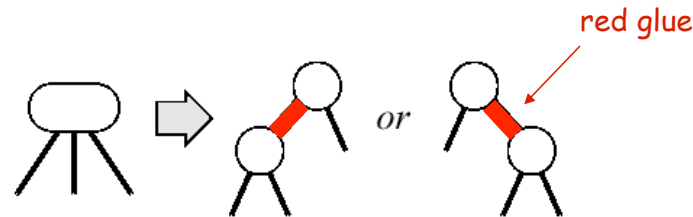
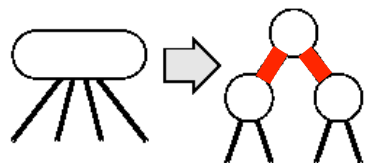
Important Properties



1. Never have two red edges in a row.
 - ▶ Red edge is “internal”, never have more than one “internal” edge in a row.
2. Every leaf has same number of *black* edges on path to root (*black-depth*)
 - ▶ Each black edge is a 2-3-4 tree edge
 - ▶ All leaves in 2-3-4 tree at same distance from root

$\implies$ path from root to deepest leaf $\leq 2 \times$ path to shallowest leaf

Important Properties



1. Never have two red edges in a row.
 - ▶ Red edge is “internal”, never have more than one “internal” edge in a row.
2. Every leaf has same number of *black* edges on path to root (*black-depth*)
 - ▶ Each black edge is a 2-3-4 tree edge
 - ▶ All leaves in 2-3-4 tree at same distance from root

$\implies$ path from root to deepest leaf $\leq 2 \times$ path to shallowest leaf

$\implies$ depth $\leq O(\log n)$

Insert

Want to insert while preserving two properties.

Insert

Want to insert while preserving two properties.

2-3-4 trees: split full nodes on way down.

Insert

Want to insert while preserving two properties.

2-3-4 trees: split full nodes on way down.

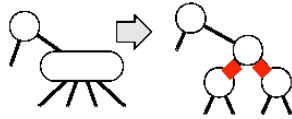
Easy cases:

Insert

Want to insert while preserving two properties.

2-3-4 trees: split full nodes on way down.

Easy cases:

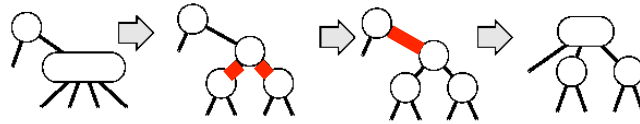


Insert

Want to insert while preserving two properties.

2-3-4 trees: split full nodes on way down.

Easy cases:

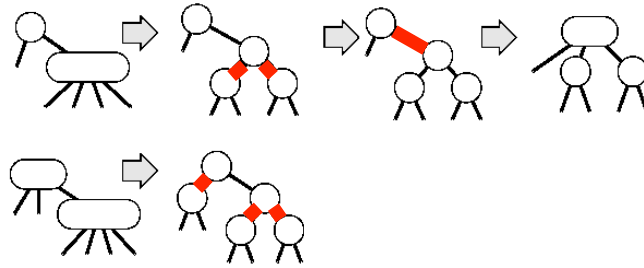


Insert

Want to insert while preserving two properties.

2-3-4 trees: split full nodes on way down.

Easy cases:

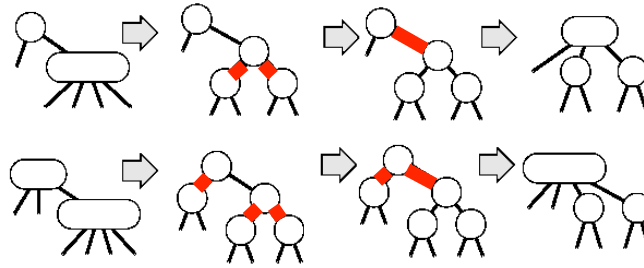


Insert

Want to insert while preserving two properties.

2-3-4 trees: split full nodes on way down.

Easy cases:

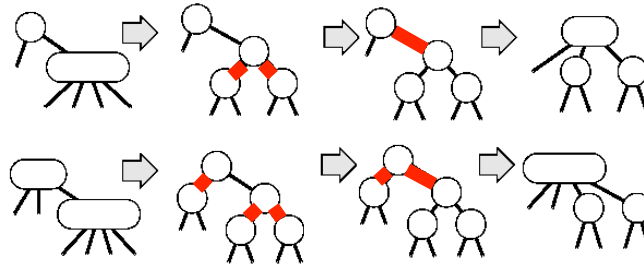


Insert

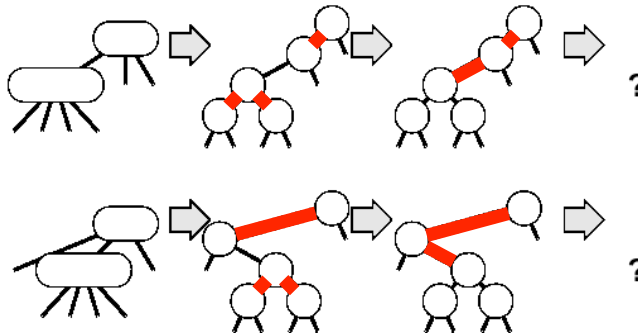
Want to insert while preserving two properties.

2-3-4 trees: split full nodes on way down.

Easy cases:



Harder cases:

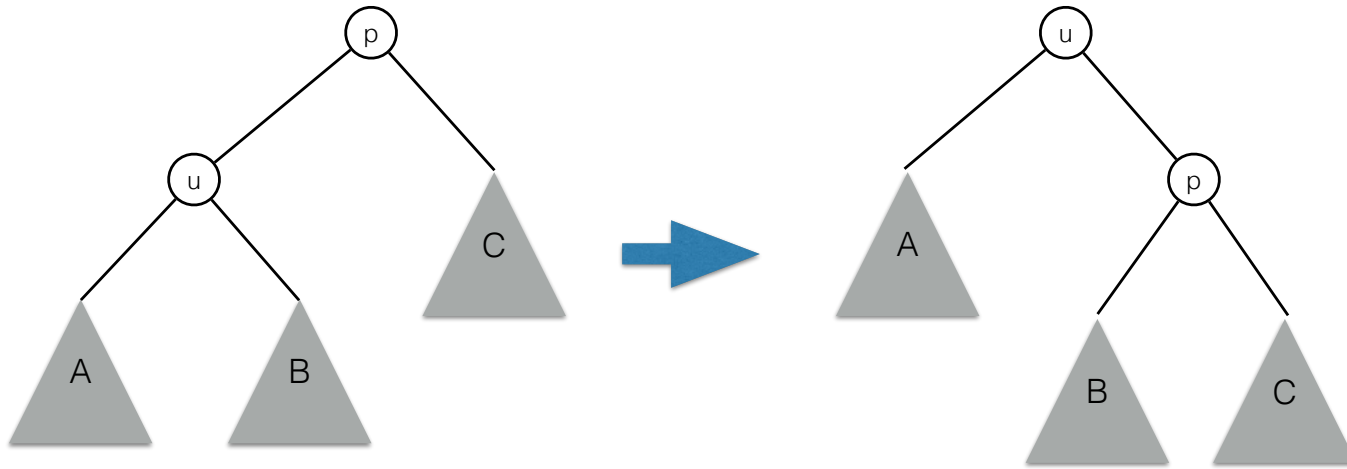


Tree Rotations

Used in many different tree constructions.

Tree Rotations

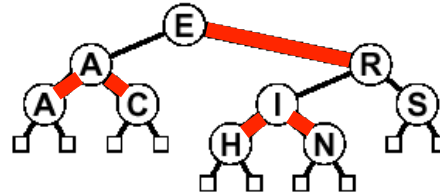
Used in many different tree constructions.



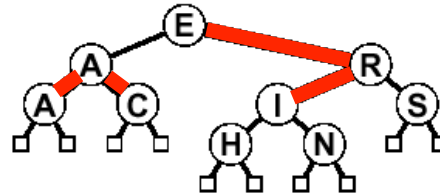
Using Rotations

Can use rotations to “fix” hard cases. Example:

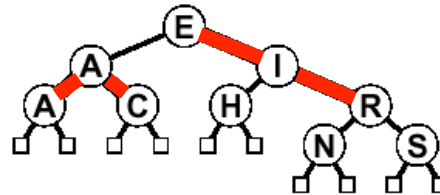
inserting *G*



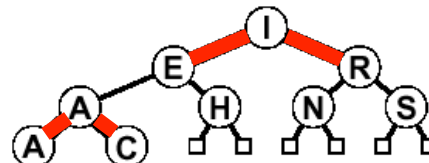
change colors



right rotate R →



left rotate E →



End

A few more complications to deal with – see lecture notes, textbook.

End

A few more complications to deal with – see lecture notes, textbook.

Main points:

- ▶ Red-Black trees can be thought of as a binary implementation of 2-3-4 trees
- ▶ Approximately balanced, so $O(\log n)$ lookup time
- ▶ Insert time (basically) same as 2-3-4 tree, so also $O(\log n)$.
- ▶ See book for direct approach (not through 2-3-4 trees).